

Robust nonlinear projection-based reduced-order models: comparative assessment of closure and manifold strategies

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Abstract. This talk presents a comparative study of ANN-enhanced projection-based reduced-order models for nonlinear parametric conservation laws. The focus is on robust online prediction under aggressive compression, where the reduced model must remain accurate after hyper-reduction and outside the small set of high-dimensional training simulations. We compare intrusive closure-based PROM-ANN models, an intrusive POD-autoencoder latent-manifold PROM, and two non-intrusive surrogates mapping parameter-time inputs either directly to POD coefficients or through a low-dimensional latent representation.

A central point of the study is consistency between data generation, training, and online evaluation. All learned models are trained from coefficient trajectories generated by a linear hyper-reduced LSPG ROM in a fixed reduced space, rather than from direct orthogonal projection of high-dimensional snapshots. The intrusive online models use the same discrete residual, Gauss-Newton LSPG solver, and ECSW hyper-reduction framework. This builds on closure-enhanced projection ideas such as PROM-ANN [1], nonlinear-manifold ROM formulations [2], hyper-reduction with ECSW [4], latent POD-DL-ROM concepts [3], and recent machine-learning-assisted PROM training and closure studies [5, 6]. We also use an M -regularized LSPG-sensitive POD metric to reduce the feedback of high-coordinate closure errors into the resolved online coordinates.

The numerical results are shown on a 250×250 two-dimensional parametric inviscid Burgers benchmark. Starting from a baseline set of nine HDM trajectories, we examine the effect of enriching the learning dataset with additional low-cost linear-HPROM trajectories, including points in an expanded parameter region. The results highlight when neural enrichment improves extrapolatory robustness, when intrusive feedback remains the limiting factor, and how closure-based and latent-manifold ROMs compare with purely non-intrusive predictors in terms of accuracy, online cost, and achieved speedup.

References

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